

scripts/autonomous-qa/release-coldstart-canary.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.AA (two-stage distribute), §6.Z.4 (pre-distribute test execution), §6.AH (containerized emulator) **Classification:** project-specific

Overview

The §6.AA / §6.Z.4 **RELEASE-variant R8 cold-start canary**.

The project has NO testBuildType=release, so the instrumentation keystone (Challenge70 / Challenge00) cannot run against the R8-minified release APK. This canary is the release-variant equivalent of the C00 cold-start gate: it boots a containerized emulator (via lib-emulator.sh, §6.AH), fresh-installs the RELEASE APK, pm clears it, cold-launches it, and confirms it SURVIVES — the exact failure class of the **1.2.19-1039 forensic anchor** (a release-only R8 painterResource crash at MainActivity.setContent that the debug build did not reproduce, and that hit every cold launch on a real device).

Anti-bluff verdict

ALL THREE must hold for PASS (a cold-start crash breaks all three):

1. NO FATAL EXCEPTION for the package in logcat after launch,
2. the app process is ALIVE (pidof) — a cold-start crash kills it,
3. the app's activity is the TOP resumed activity (dumpsys) — not just a lingering background service.

Usage

```
scripts/autonomous-qa/release-coldstart-canary.sh [--apk <path>]
[--evidence-dir <dir>]
```

Flag	Default	Meaning
<code>--apk <path></code>	<code>app/build/outputs/apk/release/app-release.apk</code>	The release APK under test
<code>--evidence-dir <dir></code>	<code>.lava-ci-evidence/1079-release-coldstart</code>	Curated evidence destination

Note the target package is `digital.vasic.lava.client` — the release variant carries NO `.dev` suffix.

Evidence

```
<evidence-dir>/
  verdict.json                # variant/package/apk_sha256/
fatal_count/pid/top_activity/verdict/note
  raw/release-coldstart-logcat.txt # full logcat dump after the
20 s settle
  raw/release-coldstart.png       # screenshot after launch
```

The APK's sha256 is recorded in `verdict.json` so the canary is pinned to the exact artifact about to be distributed (§6.Z).

Exit codes

- 0 — PASS (survived cold start, foregrounded)
- 1 — FAIL (crash / not foregrounded)
- 2 — setup error (release APK missing, emulator boot failure)

Companion files

- `scripts/autonomous-qa/lib-emulator.sh` (+ its doc) — the containerized emulator lifecycle
- `scripts/run-release-canary.sh` (+ its doc) — the sibling release-canary runner
- `docs/scripts/firebase-distribute.sh.md` — the §6.AA two-stage distribute flow this gates
- `.lava-ci-evidence/` — §6.Z distribute evidence packs